

RRC-Raptor-V4H

I2C

- User's Manual: Software

V0.0.2

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i. **VERSION LIST**

Description	Version	Date	Author
RTX_I2C	V0.0.1	2024/07/17	Jerry
RTX_I2C	V0.0.2	2024/10/29	Jerry

ii. HASHTAG

#V4H

#I2C

#BSP

1. Overview

Target：介紹 kernel configuration, 以及如何使用 I2C 工具讀寫 I2C 裝置.

2. Directory Configuration

The directory configuration is shown below.

—— drivers/i2c/busses/ —— i2c-rcar.c : I2C Driver source file

Figure 1. Directory Configuration

3. Kernel configuration

To enable the function of this module, make the following setting with Kernel Configuration. The changes affect after re-building the kernel.

```
Device Drivers --->
  *- I2C support --->
    <*> I2C device interface
    I2C Hardware Bus support --->
      <*> Renesas R-Car I2C Controller
```

Figure 2 Kernel configuration

4. Connect

提供 I2C interface 讓 User 可以自己加入 I2C device.

CN32 CAN BOARD CONNECTOR

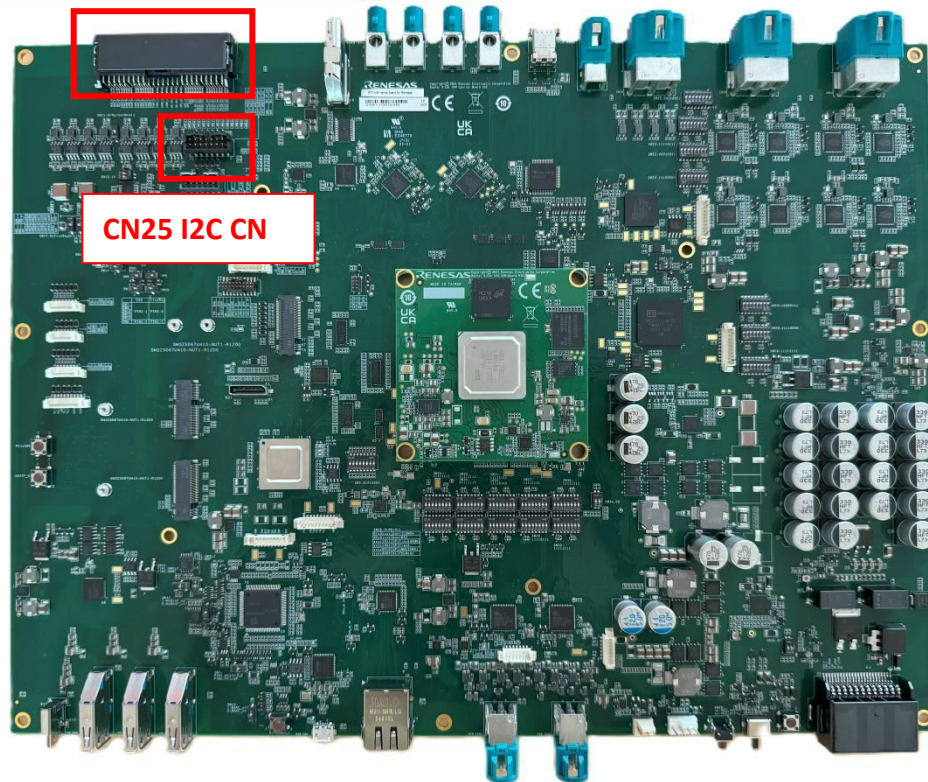


Figure 3 Connect

CN32 : CAN/I2C/GPIO CONNECTOR

CAN H / CAN L signals after passing through the internal transceiver.

PIN	SYMBOL	對應系統 bus id
1	I2C2_SDA_CNA	I2C19
2	I2C2_SCL_CNA	I2C19
3	I2C2_SDA_CNB	I2C19
4	I2C2_SCL_CNB	I2C19
5	I2C2_SDA_CNC	I2C20
6	I2C2_SCL_CNC	I2C20
7	I2C2_SDA_CND	I2C20
8	I2C2_SCL_CND	I2C20
9	I2C2_SDA_CNE	I2C21
10	I2C2_SCL_CNE	I2C21

11	I2C2_SDA_CNF	I2C21
12	I2C2_SCL_CNF	I2C21
13	D3.3V_I2C	
14	SG	
17	D5.0V1	
18	D5.0V1	

CN25 : I2C CN

I2C0~I2C4 are signals directly from the SoC.

PIN	SYMBOL	對應系統 bus id
1	D3.3V_I2C	
2	GND	
3	D3.3V_I2C	
4	GND	
5	I2C0_SCL	I2C0
6	I2C0_SDA	I2C0
7	I2C1_SCL	I2C1
8	I2C1_SDA	I2C1
9	I2C2_SCL	I2C2
10	I2C2_SDA	I2C2
11	I2C3_SCL	I2C3
12	I2C3_SDA	I2C3
13	I2C4_SCL	I2C4
14	I2C4_SDA	I2C4

5. Option Setting

Please set I2C transfer speed 1000000 or 400000 or 100000 to "clock-frequency" of device tree file in arch/arm64/boot/dts/renesas directory. No setting means 100000 Hz.

```
&i2c0 {  
    status = "okay";  
    clock-frequency = <100000>;  
    ...  
}  
  
&i2c1 {  
    status = "okay";  
    clock-frequency = <400000>;  
    ...  
}
```

I2C0, I2C1, I2C2, I2C3, I2C4, and I2C5 have Open drain buffer.

Their IntDelay are set 110ns in arch/arm64/boot/dts/renesas/ r8a779g0.dtsi

For fast mode plus, please set "i2c-scl-rising-time-ns" to 120 or lower (according to I2C specification).

6. I2C Tool

Use 可以透過 CN32(I2C2_CNA~ I2C2_CNF) 或 CN25 (I2C0~ I2C4) 來連接

Use i2ctransfer can read / write i2c device.

- I. Here is an example of Raptor connecting to I2C AT24C256 EEPROM (addr 0x50) through CN25 的 I2C1.

以下範例為

1. 對 I2C1 的 0x50 裝置讀取 0x00 0x00 位置上的 4 個 byte 的值.
2. 對 I2C1 的 0x50 裝置寫入 0x23 到 0x00 0x01 位置.
3. 再對 I2C1 的 0x50 裝置讀取 0x00 0x00 位置上的 4 個 byte 的值.發現 0x00 0x01 位置的值已變成 0x23.

```
root@v4x:~# i2ctransfer -f -y 1 w2@0x50 0x00 0x00 r4  
0xff 0xff 0xff 0xff  
root@v4x:~# i2ctransfer -f -y 1 w3@0x50 0x00 0x01 0x23  
root@v4x:~# i2ctransfer -f -y 1 w2@0x50 0x00 0x00 r4  
0xff 0x23 0xff 0xff
```

Figure 4 example of i2ctransfer method

- II. 如果要透過 CN32 的 I2CA 接上 EEPROM (addr 0x50)
將上面範例的 transfer 指令中 1 (bus 1)改為 19 (bus 19)即可.

I2ctransfer -f -y 19 w2@0x50 0x00 0x00 r4

I2ctransfer -f -y 19 w3@0x50 0x00 0x01 0x23

I2ctransfer -f -y 19 w2@0x50 0x00 0x00 r4

7. APPENDIX